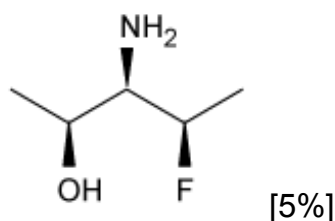
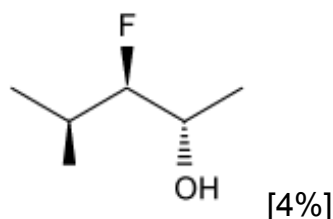


Which of the following molecular structures has the most hydrophobic character?

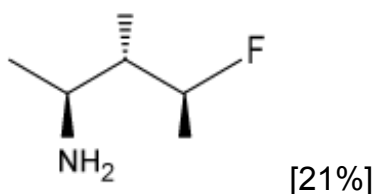
☐ A.



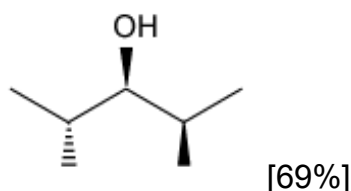
☐ B.



☐ C.



✓ ☒ D.

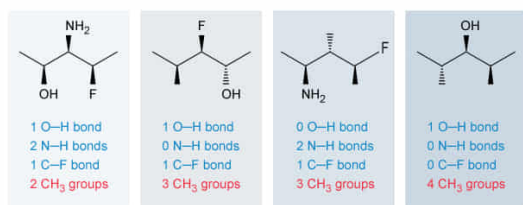


Correct

68% answered correctly

Explanation:

Assessing hydrophobic character of compounds



Increasing hydrophobic character

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Hydrophilic bonds

Hydrophobic bonds

In the structure of a water molecule, the large **electronegativity** difference between the covalently bonded H and O atoms produces a **polar covalent bond** in which the H atom has a partial positive charge and the O atom has a partial negative charge. The partial charges associated with the **dipole** of the O—H bonds cause water to readily **interact** with polar bonds or ionic charges in other molecules via **ion-dipole interactions**, dipole-dipole interactions, or **hydrogen bonding** (strong dipolar interactions from bonds involving H, N, O, and F).

Molecules with many polar bonds that promote dipolar interactions with water are **hydrophilic**

("water-loving"), whereas those with mostly nonpolar bonds that lack attractive dipolar interactions with water are **hydrophobic** ("water-fearing"). Frequently, a compound structure

will have both nonpolar and polar bonds that make a strict binary classification difficult. In such cases, the overall hydrophobic or hydrophilic **character** of the compound depends on the **relative number** of each **type of bond** and the strength of the dipoles present.

The compounds presented in this question each consist of a five-carbon chain and differ in the number of polar bonds and nonpolar methyl substituents. Among the four structures, the compound with the greatest number of nonpolar methyl ($-\text{CH}_3$) groups and the fewest polar bonds (**Choice D**) will interact with polar water molecules the least and have the most hydrophobic character.

(Choice A) This structure is very *hydrophilic* due to having one very polar O–H bond, two very polar N–H bonds, and one polar C–F bond, which are all capable of hydrogen bonding. The structure contains only two nonpolar methyl groups.

(Choices B and C) These structures have an intermediate hydrophobic character due to having fewer polar bonds, but neither compound has the most hydrophobic character among the answer choices. Each has only three nonpolar methyl groups.

Educational objective:

Molecules with many polar bonds that promote dipolar interactions with water are hydrophilic, whereas those with mostly nonpolar bonds that lack attractive dipolar interactions with water are hydrophobic. The overall character of a compound depends on the relative number of each type of bond present.

Time spent:0 sec

Subject:
General Chemistry

QID:402569

System:
Interactions of Chemical Substances